Edwin Hung
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Summary

Software Engineer with 5+ years of architecting distributed systems using Python and Java. Expert in data intensive design and cloud-native architecture (AWS/Kubernetes), with deep specialized experience in low-latency stream processing using Kafka and Redis. Proven track record of integrating AI/ML workflows into backend systems to drive engineering efficiency and business metrics. Adept at building scalable backend services that handle massive transaction volumes. Enthusiastic about Functional Programming and eager to transition to a Haskell environment.

Experience

OKX (Top 2 Crypto exchange) – Software Engineer, Trading | 10/2023 – 2/2025

- Architected a real-time stream processing pipeline using Kafka and the Pub/Sub pattern to ingest and process massive transaction volumes, ensuring 99.99% availability through robust error stats monitoring via Grafana.
- Implemented distributed module with OAuth 2.0 & HyperLogLog, tracking 1M+ unique daily users while reducing analytics memory usage by 99% (from GBs to KBs) using Redis
- Developed trading analytics platform using Elasticsearch and Redis serving 1M daily request.
- Using real-time stream processing pipelines with Kafka, built trading volume analytics to support scalable, low-latency data processing
- Engineered a high-throughput floor price service using Redis ZSETs, maintaining sub-millisecond accuracy during market spikes while offloading 95% of read volume from the primary database

Ebay, New York, NY – Software Engineer, Ads | 08/2022 – 10/2023

- Led the backend development of an internal AI chatbot using RAG (Retrieval-Augmented Generation) and LLMs, which improved engineering knowledge retrieval and operational efficiency by 30%
- Delivered a high-scale 2-day shipping ads feature using Java microservices, orchestrating cross-team launches, mentor junior engineers and implementing post-CPC reranker scoring; ultimately boosted revenue by $54 million.
- Engineered a local pickup recommendation feed discovery module, utilizing Python and Pandas to perform independent statistical analysis and model selection, increasing Click-Through Rates (CTR) by 80%.
- Managed CI/CD pipelines and automated testing to increase unit test branch coverage from 70% to 99%, reducing production rollback rates by 30%

American Express, New York, NY – Software Engineer II | 05/2021 – 08/2022

- Executed a GraphQL caching strategy that reduced API response times by 90%, significantly improving the user experience for high-traffic customer portals
- Designed and implemented cache for gateway service to reduce traffic from internal and external vendors by 90%, reducing cost by 100k using React, Redis, PostgreSQL
- Designed and implemented outlook calendar invite, reduced labor cost by 400k, using Spring routing, Microsoft 365

M&T Bank, Buffalo, NY – Software Engineer | 02/2020 – 05/2021

- Developed and tested web app including backend services for 2 million users, providing functionality such as bank to bank transfers, Zelle/QuickPay, accessible features, etc. using: .Net Core, Angular, Redis and Docker
- Designed, co-developed and deployed bank to bank transfer monitoring service for 4000 customer service users with compliance audit recording features using Swagger Doc, .Net Core, Spring Boot and Angular

Skills

Languages, Frameworks & Infrastructure: Java, Python, Go, Scala (Functional Programming), Spring, React, Linux, Kafka, GraphQL, CI/CD, Nifi, Elasticsearch, Redis, Kubernetes, PostgreSQL, Grafana, Flink,

Cloud & Analytics: AWS (EC2, S3, RDS), ClickHouse, Hive, A/B Testing

Education

University at Buffalo, SUNY — B.S. Computer Science, May 2019

Competitions

2nd Place, UB Hackathon: Flask bus lateness predictor (linear regression with 86% accuracy) with Numpy and Flask

1st Place, UB Blockchain Hackathon: Solidity-based decentralized medical records app